Question

An oxyacid $\mathbf{A}$ is an inorganic strong acid with a highly stable corresponding anion. $\mathbf{X}$ is an extremely unstable, explosive gas that can be prepared using $\mathbf{A}$ or its salts as substrates. Two methods of preparation are as follows:

(1) Under cooling, a 60% aqueous solution of acid $\mathbf{A}$ reacts with a pale yellow gas $\mathbf{B}$ to produce gas $\mathbf{X}$.

(2) In hydrogen fluoride at $-78°C$, the reaction of the cesium salt of $\mathbf{A}$ with $\mathrm{NF_4SbF_6}$ yields a precipitate $\mathbf{C}$ and a readily decomposable $\mathbf{D}$; $\mathbf{D}$ slowly decomposes into $\mathbf{E}$ and $\mathbf{X}$ at $0°\mathrm{C}$.

Vapor density determination experiments show that at $25°\mathrm{C}$, the density of $\mathbf{X}$ is $4.85\ \mathrm{g/L}$

A. The configuration of the corresponding anion of substance $\mathbf{A}$ is trigonal pyramidal.

B. The substance $\mathbf{B}$ contains elements from the third period.

C. The configuration of substance $\mathbf{E}$ is a trigonal pyramid

D. The anionic configuration of substance $\mathbf{C}$ is not octahedral

E. Only covalent bonds exist in material $\mathbf{D}$.

F. Substance $\mathbf{X}$ contains 3 oxygen atoms

G. The anions in substance $\mathbf{A}$ and substance $\mathbf{D}$ are not identical.

Answer

Correct Answer: C

Detailed Explanation

According to the ideal gas equation $PM = \rho RT$, with the default pressure set to 1 atmosphere, the relative molecular mass of substance $\mathbf{X}$ is calculated to be 118.65.

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The relative molecular mass of $\mathbf{X}$ is 118.65

The common pale-yellow gas $\mathbf{B}$ is likely to be F_2. Considering the highly unstable and explosive nature of substance $\mathbf{X}$, it is inferred to possess strong oxidizing properties, supporting the conclusion that $\mathbf{B}$ is F_2.

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$\mathbf{B}$ is F_2

Since oxygen-containing acid $\mathbf{A}$ reacts with F_2 to produce $\mathbf{X}$, it must contain at least O and F elements. Based on the relative molecular mass of $\mathbf{X}$, a table is constructed for possible combinations of O and F. Starting with 1 F atom and n O atoms, it is found that when $n = 4$, the corresponding relative molecular mass is 35.65, which is close to that of Cl. Given that $\mathbf{A}$ is an inorganic strong acid with a highly stable anion, it is deduced that $\mathbf{A}$ is $HClO_4$ and $\mathbf{X}$ is $FClO_4$.

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$\mathbf{A}$ is perchloric acid

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$\mathbf{X}$ is $FClO_4$

The cesium salt of $\mathbf{A}$ reacts with NF_4SbF_6 to form precipitates $\mathbf{C}$ and $\mathbf{D}$, suggesting a double displacement reaction. Therefore, $\mathbf{C}$ is $CsSbF_6$, and $\mathbf{D}$ is NF_4ClO_4. The decomposition of $\mathbf{D}$ yields $\mathbf{X}$, with the remaining molecular formula being NF_3, which is $\mathbf{E}$.

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$\mathbf{C}$ is $CsSbF_6$

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$\mathbf{D}$ is NF_4ClO_4

The anion of $\mathbf{A}$ has a tetrahedral geometry, $\mathbf{E}$ has a trigonal pyramidal geometry, and the anion of $\mathbf{C}$ has an octahedral geometry. $\mathbf{D}$ contains both covalent and ionic bonds, so option C is correct.

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The anion of $\mathbf{A}$ has a tetrahedral geometry

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$\mathbf{E}$ has a trigonal pyramidal geometry

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The anion of $\mathbf{C}$ has an octahedral geometry